

Across primates, social adversity predicts poorer health, elevated mortality, and altered reproductive function. Yet, the mechanisms that link social environments to health and reproduction, as well as the evolutionary forces shaping these sensitivities, remain poorly understood. As a biological anthropologist and geneticist, I combine long-term field anthropological data from humans and non-human primates (NHPs) with genomic analyses of tissue samples and targeted cell line experiments to investigate **how social experiences across the life course become biologically embedded and contribute to reproductive function, evolutionary fitness, and health.**

While isolating the effects of social adversity on reproduction and health in humans is complicated by confounding factors (e.g., access to health care, health behaviors), NHPs offer a powerful system to examine how and why social adversity impacts health from an evolutionary perspective. My research program therefore takes a comparative approach to studying the social determinants of health across the lifecourse, with a particular focus on reproductive function—a driver of evolutionary fitness and a critical, yet often overlooked, aspect of women's health and aging. I will do so by: 1) expanding functional genomic tools in NHPs, 2) identifying molecular pathways through which social conditions shape reproductive outcomes, and 3) uncovering the evolutionary dynamics underlying female reproductive function and disease. My work aims to identify genomic regions and physiological processes most sensitive to adversity and to contextualize reproductive aging and disease in relation to the social environment. By doing so, it provides anthropologically informed insight into the evolutionary origins of environmental sensitivity and complex traits in humans and other primates.

PAST AND CURRENT RESEARCH

Graduate research: My graduate research was motivated by the goal of understanding how genetic variation and sexual selection shape reproduction and health in NHPs. For my master's work, I explored links between sociality, reproduction, and immune function in a free-ranging NHP population using a non-invasive biomarker of inflammation. I found evidence for an immunological cost of reproduction and a behavioral mechanism for social buffering during periods of physiological stress (Petersen et al. 2021, *Behavioral Ecology and Sociobiology*), highlighting how social conditions can impose physiological costs and generate resilience. For my dissertation, I investigated the genetic architecture underlying immune function and mate choice (Petersen et al. 2022, *Ecology and Evolution*), as well as behavioral mating preferences and post-copulatory responses within the female reproductive tract in the olive baboon. This work provided the first evidence of *in vivo* sperm discrimination in a primate species (Petersen et al. 2025 *bioRxiv* in revision at *PLOS Biology*), expanding our understanding of the genetic basis of mate choice and how inter-sexual selection may contribute to infertility in humans.

Post-doc research: Motivated to deepen my genomic expertise and explore how the genome both shapes and responds to social experiences, I expanded my initial focus on mate interactions to study the effects of early life adversity. As an NSF postdoctoral fellow, I helped generate the largest comparative multi-tissue DNA methylation dataset to date from a long-term population of rhesus macaques. Using detailed information about each individual's social and ecological experiences [4-7], I found that early life adversity targets genomic regions implicated in aging, and accelerates epigenetic aging in a tissue-specific manner (Sadoughi*, Petersen* et al., 2025, *bioRxiv* in revision at *Science*). In this same macaque study population, I also found that early life environments shape life history strategies, altering patterns of both reproduction and growth (Petersen et al., 2025, *bioRxiv* in revision at *PNAS*). Additionally and importantly, I expanded my skillset beyond work with NHPs to include 1) demonstrations of early life effects on later life cardiometabolic health in a long-term subsistence-level human population (Petersen et al., 2025, *bioRxiv* in revision at *Proc R Soc B*), and, 2) I led the development of novel wet-lab methods to test causal effects of DNA methylation on gene expression, a key hypothesized mechanism through which social experiences across the life course can generate lasting changes in physiology (Petersen et al., 2025, *Genome Research*).

I have a strong record of securing competitive funding from diverse sources. My dissertation research was supported by major biological anthropology grants, including an NSF Doctoral Dissertation Improvement Grant, as well as awards from the Leakey Foundation and Wenner-Gren Foundation. I also

obtained multiple society-funded grants (e.g., Animal Behavior Society, American Society of Mammalogy, Society for Integrative and Comparative Biology) and career-advancing fellowships, including the NSF SBE Postdoctoral Research Fellowship. My current work, with its biomedical applications, is funded by the NIH's National Institute on Aging.

FUTURE RESEARCH PLANS

My future lab will integrate theories and questions from biological anthropology with methods in functional genomics. I will leverage three complementary systems: free-ranging NHP populations, biobanked tissues, and in vitro cell line experiments. During my first years as a tenure-track scientist, I will advance the following projects:

Social and environmental factors contributing to reproductive aging

Primates show striking variation in reproductive aging, including the timing and extent of reproductive senescence—a phenomenon increasingly recognized as widespread across primates. While ultimate explanations suggest that reproductive senescence may confer inclusive fitness benefits, proximate factors—including early life adversity, energetic trade-offs, and hormonal dynamics—likely shape individual reproductive trajectories and influence the evolution of reproductive aging. Through my ongoing collaboration with the Cayo Biobank Research Unit (CBRU) and Dr. Noah Snyder-Mackler (Arizona State University), I am investigating the molecular mechanisms that contribute to female reproductive senescence in humans and NHPs. Using a novel cost-effective method for targeted DNAm sequencing that I helped to benchmark [12], I will generate DNAm profiles from the reproductive tissues of ~100 longitudinally monitored female rhesus macaques and use statistical modeling to identify adversity associated genomic regions and build epigenetic clocks—a molecular tool that can estimate a tissue's "biological age". Because deviations between chronological age and biological age are strongly linked to health trajectories, these clocks provide a powerful lens into the pace of aging. I will use these epigenetic clocks in conjunction with long-term demographic and life history information to test how the socioecological environment shapes the pace of aging in female reproductive tissues, evaluating whether adversity accelerates reproductive aging and offering a potential explanation for variation in the timing of reproductive senescence between individuals. In humans, female reproductive tissues reach the end of their functional lifespan far earlier than most other tissues, producing a menopause that is unusual among primates. To investigate whether tissue-specific differences in epigenetic aging underlie this discrepancy between humans and other NHPs, I will also compare the pace of aging in reproductive versus non-reproductive tissues in female rhesus macaques drawing upon a previously published DNAm dataset from 12 non-reproductive tissues originating from the same individuals [8]. To compare to humans, I will similarly build a cross-tissue epigenetic clock using the publicly available GTEx resource and evaluate whether reproductive tissues age more rapidly compared to other tissues in humans compared to rhesus macaques. This work will clarify how reproductive tissues diverge from other organ systems and provide insight into the social drivers of reproductive senescence and the evolutionary origins of human menopause.

Linking social adversity to women's reproductive health through the epigenome

Social experiences profoundly shape complex traits, including the risk and progression of reproductive diseases such as endometriosis. Epigenetic mechanisms, particularly DNAm, likely mediate these effects. Yet, research in reproductive tissues has been constrained to candidate-gene or reduced-representation approaches, with almost no direct evaluation of how social experiences shape epigenetic variation across the entire genome. This leaves a critical gap in our understanding of genome-wide regulatory pathways connecting social exposures to complex phenotypes, such as disease. To address this gap, I will generate the first comprehensive map of genome-wide DNAm in endometrial tissue from women with and without endometriosis and integrate these profiles with macaque data generated in Study #1 to test whether social adversity modifies DNAm at disease-relevant regulatory regions contributing to gene expression. This project offers a unique opportunity to directly connect the socioecological environment to disease-relevant molecular pathways, empirically testing for links between **social environment** → **epigenome** → **gene expression** → **complex trait**. Furthermore, by situating

these findings within an anthropological framework, this work will link molecular mechanisms to the broader social and ecological contexts in which reproduction evolves. My established collaborations with Drs. Digna Velez Edwards and Emily Hodges at Vanderbilt University Medical Center, and with Kerry Wiles at the Cooperative Human Tissue Network at Vanderbilt, position me to carry out this work. Beyond producing the first high-resolution reference of epigenetically regulated loci relevant to endometriosis, this project establishes a broadly applicable framework for linking social experiences to complex traits across species and bridges proximate biological processes with ultimate anthropological questions, such as why reproductive systems are particularly sensitive to social conditions, how social environments become biologically embedded through evolutionarily conserved regulatory pathways, and why complex reproductive diseases persist in human populations despite their apparent fitness costs.

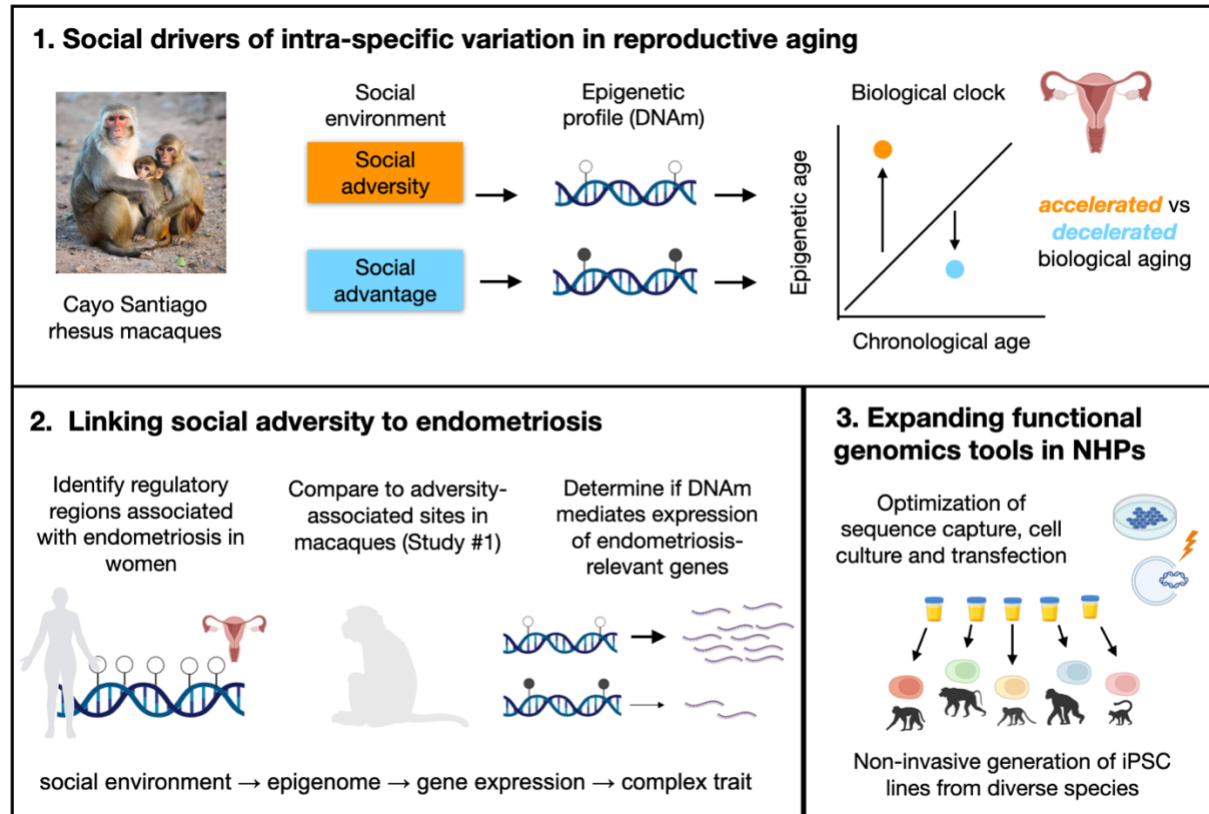
Expanding functional genomic tools in lesser studied primate species and tissues

A major barrier in primate functional genomics is the lack of species- and tissue-specific resources, which limits our ability to understand how gene regulation evolves and contributes to complex traits across the primate lineage. To address this, I am currently optimizing sequence capture methods for use in massively parallel reporter assays (MPRAs), a wet-lab method which identifies regulatory genomic sequences but has been limited in its application across species due to challenges in cell culture, transfection, and sequence capture methods, which have generally been optimized in humans. Adapting MPRAs for NHPs will allow us to annotate regulatory sequences across species, providing a comparative framework to study the evolution of gene regulation. In the near term, my lab will refine MPRA methods for use in NHPs by improving cell culture and transfection protocols, with the long-term goal of expanding the availability of NHP cell line models using induced pluripotent stem cells (iPSCs). While iPSCs are often derived from blood or tissue, recent advances in non-invasive iPSC generation could greatly expand their use across difficult to access species and cell types. To advance these methods, I will collaborate with Dr. Genevieve Housman (Max Planck Institute for Evolutionary Anthropology), who has validated protocols for generating and differentiating NHP iPSCs. Through my training in the New York Consortium in Evolutionary Primatology (NYCEP) and my active collaborations with multiple NHP field sites, I have developed extensive opportunities to partner with colleagues across institutions to access blood, tissues, or non-invasively collected samples as starting material for iPSCs from diverse primate species. Together, these resources will enable my lab to investigate how environmental exposures influence gene regulation across species and cell types, providing a comparative framework to understand the evolution of gene regulation and how social and environmental variation shapes molecular phenotypes across primates.

SUMMARY

My lab will investigate the molecular pathways and evolutionary forces through which social experiences become biologically embedded and influence complex traits—combining NHP field work with analyses of biobanked tissues and experimental work in cell lines. My expertise in evolutionary genomics, functional epigenetics, and primate behavioral ecology, combined with my extensive collaborative network, provides a strong foundation for this proposed integrative work. To do this work, I would be prepared and excited to build out a wet lab genomics space—a fantastic resource for undergraduate and graduate level training—however it is not essential if such resources are unavailable. The breadth and flexibility of my research program not only allows me to excel in different research environments, but also positions me to compete for diverse funding sources, including NSF, Leakey, Wenner-Gren, and NIH. I look forward to building collaborations with faculty in the Department of Anthropology at the University of New Mexico. In particular, Drs. Martin Muller and Melissa Emery Thompson, whose expertise in primate behavioral ecology and reproductive endocrinology, would greatly enrich the ecological and evolutionary interpretation of my genomics findings. I am also eager to continue collaborating with Dr. Ian Wallace, with whom I worked during my postdoc to investigate environmental determinants of health in the Orang Asli of Peninsular Malaysia. His expertise in evolutionary medicine and human evolution will provide a valuable perspective as I extend my research to understand human health through an evolutionary lens, and would be helpful as I foster future collaborations involving human populations. Together, my program will contextualize reproductive physiology by connecting social dynamics with genomic processes, advancing both human and NHP biology and its applications to the study of human evolution.

Figure 1. Overview of future research plans. In study #1 & 2, filled lollipops represent the addition of a methyl group to DNA while open lollipops represent no methylation.



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